June 4, 2025

Submitted Via Electronic Delivery to Regulations.gov

The Honorable Mehmet Oz
Administrator
Centers for Medicare & Medicaid Services
Department of Health & Human Services
Hubert H. Humphrey Building
200 Independence Ave SW
Washington, DC 20201

Re: Request for Information; Health Technology Ecosystem

Dear Administrator Oz:

The Sparx Group, Inc. (Sparx) appreciates the opportunity to provide public comments on this request for information (RFI) issued by the Centers for Medicare and Medicaid Services (CMS) relating to health information technology infrastructure and approaches that CMS could take to better harness the power of technological and data innovations to improve the Medicare and Medicaid programs and care delivered to beneficiaries. In these public comments, we focus on ways in which CMS could utilize technologies based on predictive algorithms to enhance program integrity efforts in the Medicare and Medicaid programs.

Sparx is a Risk Operating System (RiskOS) that was founded to support risk-bearing health care entities in the Medicare program. Sparx currently supports over 20 percent of participants in the Accountable Care Organization (ACO) Realizing Equity, Access, and Community Health (REACH) model and a growing proportion of participants in the Medicare Shared Savings Program (MSSP), the Kidney Care Choices (KCC) model, the Transforming Episode Accountability Model (TEAM), and the Guiding an Improved Dementia Experience (GUIDE) model. Sparx currently supports these Medicare-contracting entities in identifying anomalous claims through advanced data analytics and feedback from industry stakeholders.

Sparx was founded as a healthcare technology and analytics company that leverages data science, actuarial, and artificial intelligence expertise to transform industry processes. The initial products launched by Sparx focused on capital and risk management, such as financial guarantees and stop-loss. Sparx helps customers analyze these risks through their analytics platform and access to 100% of Medicare Fee-For-Service (FFS) claims data. The Sparx guiding philosophy is that legacy healthcare processes require significant modernization, particularly through scalable technology solutions that empower Medicare providers, payers, and patients with actionable intelligence and seamless data integration.

The questions outlined in this RFI reflect a forward-looking vision for CMS to strengthen the Medicare program through enhanced data transparency, interoperability, patient empowerment,

and value-based care (VBC) alignment. Specifically, multiple RFI items emphasize the critical role that real-time data access, technology integration, and actionable insights play in advancing both patient outcomes and program integrity. These themes are linked to fraud, waste, and abuse (FWA) prevention.

CMS's ability to deliver effective care navigation, digital health tools, and longitudinal patient records is directly tied to the accuracy, completeness, and security of its underlying data. Yet many developers, providers, and value-based care (VBC) enablers lack consistent access to timely claims, clinical, and pricing data, limiting their capacity to deliver patient-centered innovations. This fragmentation also weakens oversight capabilities, allowing billing irregularities, improper payments, and fraudulent schemes to go undetected until long after payment has occurred.

VBC programs depend on accurate attribution, risk stratification, and quality measurement (in particular VB-4, VB-12, VB-13). By prioritizing FWA detection technologies and enforcing standards for data standardization and claims surveillance, CMS can ensure that payment innovations do not inadvertently introduce new vectors for fraud. This is particularly urgent in environments with fragmented capitation rules, asymmetric payer-provider contracts, and lack of visibility across care settings.

In this context, a modernized, AI-enabled fraud surveillance infrastructure, such as the one proposed in these comments, can reduce improper payments, protect trust fund solvency, and reinforce the credibility of CMS's digital and value-based care strategies.

Proposed Approach

To address FWA, a multi-pronged approach is needed to support CMS in prevention and mitigation before FWA payments are made to Medicare providers and suppliers. This multi-pronged approach should incorporate the latest available claims information from different sources such as CMS and the industry, advanced analytics techniques such as artificial intelligence and machine learning, and continuous feedback loops. After analyzing all of this information, outputs can be bucketed into the provider level (i.e., is a particular provider submitting fraudulent claims?) and claim level (i.e., is a particular claim submitted to Medicare fraudulent?). As an example, we collaborated with an ACO to identify a major skin substitute fraud scheme approximately 18 months before national headlines, underscoring ACOs' role as the system's early warning system. Also, we are actively working with ACOs on patients who are susceptible to being charged highly expensive skin substitutes.

After certain providers and/or claims are flagged as suspected fraud, CMS can establish a rapid response infrastructure designed to mitigate FWA before additional improper payments are made. More specifically, CMS can take the outputs of this analysis and can work with claims payments partners to escrow claims related to active investigations. The rapid response process can be completed within 30 days and prevent significant areas of fraud from emerging within the Medicare program.

Sparx

Inputs into Process

A FWA platform can be built on four unique and complementary data pipelines that can support CMS in a holistic and robust FWA surveillance system. These data sources are:

1. 100% of Medicare FFS claims data
2. Claim and Claim Line Feed (CCLF) claims data
3. Industry reporting channels
4. Non-claims-based CMS Program Data

The below section describes each data set and provides more detail on the importance of including each data set in the FWA monitoring system.

100% of Medicare FFS Claims Data

The CMS claims dataset is important for understanding what claims have been paid by CMS. This data set includes medical and pharmacy claims, which are both areas where suspected FWA has occurred in recent years. As examples, we saw elevated billing to Medicare in telemedicine and COVID testing in 2020-2021, urinary catheters and ostomy bags in 2022-2023, and skin substitutes, biologics, graft materials and genetic testing in 2024-present. Also, this data set is national and captures fraud “hot spots” where multiple providers submit fraudulent claims as well as regional and national data that can be used to develop benchmarks for normal claims activity.

CMS and Medicare Administrative Contractors (MACs) regularly process more than one billion Medicare FFS claims each year. However, MAC prepayment processes do not account for emerging trends outside of their geography or claim subtype. As a result, additional information could be shared throughout the Medicare payment system to identify and stop more FWA before occurring.

The FWA monitoring system can incorporate the latest available claims data aggregated by CMS. Typically, CMS aggregates this data set monthly. For example, claims that were incurred, processed and paid before May 30, 2025 are included in this system by mid June 2025. Having visibility into near real-time claims payment patterns is critical in analyzing emerging FWA and identifying anomalous patterns before these claims grow into significant payment amounts. Unlike sampled or aggregated data sources, the 100% FFS data set will allow for granular provider-level and claim-level analytics across all service types and enable AI / ML models to detect patterns in utilization that differ from expectations.

Claim and Claim Line Feed (CCLF) claims data

Entities that partner with ACOs in the Medicare program maintain extensive internal data sets, typically comprising over 1 million member months from Medicare ACOs and provider claims. This data set, based on the CCLF, includes detailed Medicare ACO attribution data, patient

alignment, and historical utilization for Medicare beneficiaries who participate in VBC arrangements.

CMS can cross-validate the 100% of Medicare claims dataset to increase the precision of anomaly detection models.

Industry reporting channels

Similar to above, entities that maintain a direct channel with Medicare ACOs aggregate valuable information about early detection of high-dollar or rapidly changing claims that can be indicative of FWA. These entities can provide structured and unstructured claims alerts based on their internal analytics.

This data source can serve as another source for cross-validation of the 100% of Medicare claims dataset. In particular, data organized by Medicare ACO and industry stakeholders can inform CMS of risk concentration among ACOs, by geography, and by specialty.

Non-Claims-Based CMS Program Data

Based on other available information published by CMS, entities with experience in this area can complete real-time assessments of suspect provider data and flagged claims. Discussions with industry stakeholders, including those who are involved in underwriting or compliance monitoring functions, can serve as early warning indicators for fraudulent claims. Incorporating alerts or aggregate risk signals from these Medicare partners can strengthen CMS's ability to detect and respond to emerging fraud patterns more quickly than through traditional claims adjudication pathways on its own.

Taken together, these four data sources enable CMS to switch from lagged retrospective detection into near real-time prevention of FWA.

FWA Algorithm

The four data sources described above can be used to flag suspicious claims on a real-time basis and create actionable insights for CMS. A two-pronged approach can be used for this purpose by flagging:

1. Suspected fraudulent providers by NPI
2. Suspected fraudulent claims

The outputs from these models can be continually refined by a team of qualified Subject Matter Experts (SMEs) such as health actuaries, clinicians, and former MAC operators to enhance real-time reinforcement learning and clinical validation.

The below section describes each algorithm and the reason why each data set should be included in the FWA monitoring system.

Suspected fraudulent providers by NPI

For the first model, CMS can engage a partner to perform various Artificial Intelligence and Machine Learning (AI / ML) techniques to flag suspected fraudulent behavior. AI and ML techniques can be performed at varying levels of grouping, maximizing the chances of identifying unusual billing patterns. Examples of groupings should sufficiently capture how and where providers perform services and bill services to Medicare. These AI and ML techniques can be performed by a cross-functional team of SMEs and can employ the latest available technology to support the analysis.

Suspected fraudulent claims

For the second model, CMS can use Unified Medical Language System (UMLS) standards to map ICD-10 to HCPCS codes and evaluate clinical appropriateness, financial outliers, and policy violations (e.g., LCD/NCD misalignment, SAHS-specific edits). This unsupervised model combines gradient boosting with autoencoders to produce a fraud likelihood score at the claim level, based on learned deviations from legitimate diagnostic and billing behavior.

Reinforcement learning

As illustrated by how quickly fraudsters have operated over the last several years, including claims such as telemedicine, COVID testing, urinary catheters, ostomy bags, skin substitutes / biologics / graft materials, and genetic testing, among others, any successful CMS FWA monitoring solution should include reinforcement learning. A cross-functional team of SMEs, including health actuaries, credentialed clinicians, and former MAC operators, can assess algorithmic outputs for accuracy and real-world fidelity. Also, CMS can work with this type of external partner to further validate their findings with on-the-ground intelligence from the industry, including through direct contact with Medicare ACOs and industry stakeholders. Feedback from these various sources would allow CMS and their partners to continuously inform model retraining and parameter refinement.

How CMS can use these outputs

After performing an assessment and flagging a provider and/or claim, CMS can enable a rapid response process including escrow coordination and public oversight to stop fraudulent claims.

Discrete steps that can be taken include:

Escrow Process

1. MACs would escrow suspect payments during active investigation (typically within a 30-day resolution window).

2. Final determinations would be reported to the CMS Fraud War Room, the HHS OIG, and the Fraud Data Operations Center.
3. Escrowed claims would then be either released or reversed in accordance with CMS policy.

Public Oversight

4. In parallel, a vendor could maintain a publicly accessible dashboard that publishes the codes, provider types, and geographies associated with active investigations, promoting transparency and early awareness among stakeholders.

Stakeholder Empowerment

5. A FWA reporting system can enable Medicare stakeholders such as ACOs to take proactive action to prevent exposure to suspected fraudulent schemes. Examples include patient referral redirection and patient education. These steps directly relate to questions in this RFI, including PC-3 and PC-8.

Conclusion

To effectively support CMS in advancing FWA prevention and strengthening Medicare's data infrastructure, the selected technical changes and new partnerships coming from this RFI should bring together a multidisciplinary team of SMEs, specifically health actuaries, clinicians, and former Medicare program operators. To maximize effectiveness of identifying FWA, the partners should have established relationships across the healthcare ecosystem, including with provider organizations, payer networks, and other industry stakeholders. These connections will enable timely intelligence-sharing, reinforce model accuracy, and ensure that fraud detection efforts are grounded in operational realities.

Working with entities that have deep expertise and broad industry reach will be essential to achieving CMS's goals and the current administration's goals of proactive oversight, program integrity, and sustainable innovation. We appreciate CMS' interest in better leveraging technology to improve health care and enhance program integrity in Medicare and Medicaid. Please do not hesitate to contact us if you have any questions.

Sincerely,

Ricky Pati
Co-Founder and Chief Executive Officer
Sparx Group, Inc.